The L_0 dichotomy

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Abstract

Based on Bernshteyn's short proof of the G_0 dichotomy [Ber24], I give a short proof of the L_0 dichotomy [CMSV].

1 Introduction

Throughout this entire work, G denotes an analytic graph on a Polish space $V(G) = X$. That is, X is a separable completely metrizable topological space, and G is a symmetric and irreflexive relation on X which, as a subspace of $X \times X$ is the continuous image of a Polish space.

I denote the Borel chromatic number of G by $\chi_B(G)$, that is, the least number of colours k needed to find a mapping $c : X \to k$ such a way that adjacent vertices in G are given different colours. I employ the symbol $H \to_c G$ (respectively, $H \to_B G$) to abbreviate the situation where there is a continuous (respectively Borel) homomorphism from the analytic graph H to G. It is straightforward to show the following.

Fact 1.1. *If $H \to_c G$ or, more generally if $H \to_B G$, then $\chi_B(H) \leq \chi_B(G)$.*

Using a standard greedy algorithm argument, it is clear that finite graphs G of maximum degree Δ satisfy $\chi(G) \leq \Delta + 1$. Interestingly, a similar fact holds for Borel graphs of uniformly bounded degree.

Theorem 1.1 (Proposition 4.6 in [KST99])**.** *If G is a Borel graph on a Polish space all of whose degrees are bounded by the natural number k, then $\chi_B(G) \leq k + 1$.*

2 Finite approximations

Fix the parameter $c \in \omega^\omega$ and a path of length ω on the vertex set $\{p_n\}_{n<\omega}$. That is, $\{(p_n, p_{n+1}) : n < \omega\}$. Naturally, I could assume that $p_n = n$, but this labeling will aid distinguish it from other indices.

I recursively construct a family of graphs $\{L_n^c : n < \omega\}$ such that for all $n < \omega$, the following properties hold. Take note of the fact that to construct up to L_n^c, I only require knowledge of $c \restriction n$.

(i) L_n^c is a finite path.

(ii) If $n > 0$, the endpoints of L_n^c are $e_i^n := (p_0)^\frown(0)^{n-1}{}^\frown(i)$, for $i < 2$.

(iii) Every vertex of L_n^c is of the form $(p_k)^\frown t$ for some k and $t \in 2^{\leq n}$. Unless $t = \emptyset$, I say the vertex is a *non-path* vertex.

Let $L_0^c = \{(p_0, \ldots, p_{c(0)})\}$ and suppose that L_n^c has been constructed. L_{n+1}^c is a path obtained from joining two copies of L_n^c by a path of length $c(n)$ (meaning, $c(n)$ *edges*), but I must update the vertex labels in order to distinguish the two copies. I do this by simply appending 0 to each vertex of the first copy, and 1 to the other. Formally, I start by adding the vertex $v^\frown(i)$ for every $i < 2$ and $v \in L_n^c$. Adding the path $((e_1^n)^\frown(0), p_0, p_1, \ldots, p_{c(n)}, (e_1^n)^\frown(1))$ completes the construction of L_{n+1}^c, and its endpoints are

$$e_i^{n+1} := (e_0^n)^\frown(i) = (p_0)^\frown(0)^n{}^\frown(i),$$

for $i < 2$. This finished the recursion.

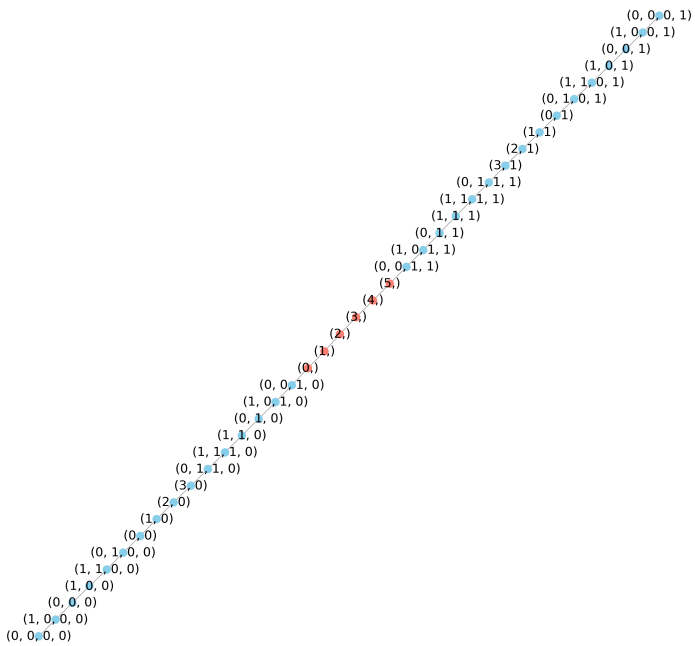
Understanding the intuition behind what (iii) says about a vertex v in L_n^c, the number $|v| - 1$ indicates that v was added that many "recursive stages ago" at the position of p_k in the path joining the two copies of the previous paths; the sequence t tells the story, in order, of which of the two copies of the previous path the vertex lies in. In other words, vertices of the form $(p_k)^\frown t$ for $t \in 2^n$ are precisely those that were copied from the original L_0^c in all possible positions along L_n^c, doubling the number of copies at each stage, for example.

Based on our discussion, a vertex in some L_n^c is completely determined by the stage $n - m$ at which it was added in the position of p_k and then copied according to t.

3 A notion of smallness for copies

By definition, the set of edges of G is the continuous image of some Polish space E, let us say under π. For a finite graph H, consider $\mathrm{Hom}(H, G)$ the set of all maps

Figure 1: L_3^c where $c(n) = 2n - 1$.



$$\varphi : V(H) \to V(G)$$
$$\varphi : E(H) \to E$$

that satisfy that for all $uv \in E(H)$, $\pi(\varphi(uv)) = \varphi(u)\varphi(v)$. Formally, φ is really two maps, however I assume that $V(H) \cap E(H) = \emptyset$ and so there is never any confusion. Notice that $\mathrm{Hom}(H, G)$ is a Polish space, being a closed subset of the product space $X^{V(H)} \times (E)^{E(H)}$.

Given $\mathcal{H} \subseteq \mathrm{Hom}(H, G)$, I define $\mathcal{H}(u) := \{\varphi(u) : \varphi \in \mathcal{H}\}$, which is analytic whenever $\mathcal{H}$ is Borel. Also, $\mathcal{H}(uv) := \{\varphi(uv) : \varphi \in \mathcal{H}\}$.

Definition 3.1.

1. Let $k < \omega$. I say an analytic set $A \subseteq V(G)$ has property $\Phi(A, k)$ if all odd walks of G with endpoints in A have length at most $2k - 1$.

2. For, $n > 0$, a Borel set $\mathcal{L} \subseteq \mathrm{Hom}(L_n^c, G)$ is called *tiny* if there is a non-path vertex $u \in V(L_n^c)$ and a natural k such that $\Phi(\mathcal{L}(u), k)$. For $n = 0$, L_n^c is just an edge consisting of the two path vertices p_0 and p_1, so in this case I define $\mathcal{L} \subseteq \mathrm{Hom}(L_0^c, G)$ as *tiny* if there is a natural k such that $\Phi(\mathcal{L}(p_1), k)$.

3. A set $\mathcal{L} \subseteq \mathrm{Hom}(L, G)$ is *small* if it is the countable union of tiny sets. This notion defines a σ-ideal.

4. A set is *large* if it is not small. Notice that when H is the graph of one single vertex, I can define $\mathrm{Hom}(H, G)$ to be $V(G)$ and then for any $\mathcal{H} \subseteq \mathrm{Hom}(\cdot, G)$, $\mathcal{H}(\cdot)$ is essentially the same as $\mathcal{H}$.

I need the following auxiliary result that links the 2-colourability and countable chromaticity of a graph in this context.

Lemma 3.1 (Claim 3.5 of [CMSV])**.** *If $\Phi(A, n)$, then there is an invariant (that is, union of connected components) Borel set $B \supseteq [A]_{E_G}$ that induces a 2-Borel-colourable subgraph of G, that is, there is a Borel proper colouring $c_B : G \restriction B \to 2$.*

Theorem 3.1. *If $\chi_B(G) > 2$ then $V(G)$ is large.*

Proof of theorem. Suppose that $V(G) = \bigcup_{m<\omega} \mathcal{H}_m$ where each $\mathcal{H}_m$ is a tiny Borel set. By Lemma 3.1, there is a G-invariant Borel set $B_m \supseteq [\mathcal{H}_m]_{E_G}$ with a Borel 2-coloring c_m. Define $c : V(G) \to 2$ by letting $m(x) := \min\{m < \omega : x \in B_m\}$ and $c(x) = c_{m(x)}(x)$. Then c is a Borel 2-colouring because each B_m is G-invariant. $\square$

Since L^c_{n+1} is obtained from two copies of L^c_n joined by a path, I require a way to refer to the distinct copies of L^c_n inside of L^c_{n+1}. By (iii) above, for every $\varphi \in \mathrm{Hom}(L^c_{n+1}, G)$, I define $\varphi^i \in \mathrm{Hom}(L^c_n, G)$ on every non-path vertex u of L^c_n by $\varphi^i(u) = \varphi(u^\frown(i))$ for all $u \in V(L^c_n)$ and $\varphi^i(uv) = \varphi(u^\frown(i), v^\frown(i))$ for all $(u, v) \in E(L^c_n)$. Thus, given $\mathcal{L} \subseteq \mathrm{Hom}(L^c_n, G)$, define $\mathcal{L}^{+c}$ as the set of all $\varphi \in \mathrm{Hom}(L^c_{n+1}, G)$ for which $\varphi^i \in \mathcal{L}$ for both $i < 2$. Notice that if $\mathcal{L}$ is Borel, then so is $\mathcal{L}^{+c}$.

The length of the path I add needs to be updated dynamically throughout the proof. Formally, I may need to adjust the parameter c by moving into a different branch of the Baire space when constructing the graphs L^c_n. Notice that, for two reals $c, d \in \omega^\omega$, if $d \restriction n = c \restriction n$, then $L^c_k = L^d_k$ for all $k < n$.

Lemma 3.2. *Let $n > 0$.*

(a) Suppose that c only takes odd values, that k is a natural and $t \in 2^{<n}$ is such that $(p_k)^\frown t^\frown(i)$, for $i < 2$, are non-path vertices of L^c_n (see (iii) above). Then, they are an odd distance apart in L^c_n.

(b) If $\mathcal{L} \subseteq \mathrm{Hom}(L^c_n, G)$ is a large Borel set, then there exists $d \in \omega^\omega$ such that $d \restriction n = c \restriction n$ and $\mathcal{L}^{+d}$ is a nonempty subset of $\mathrm{Hom}(L^d_{n+1}, G)$.

(c) If $\mathcal{L} \subseteq \mathrm{Hom}(L^c_n, G)$ is a large Borel set, then there exists $d \in \omega^\omega$ such that $d \restriction n = c \restriction n$ and $\mathcal{L}^{+d}$ is a large subset of $\mathrm{Hom}(L^d_{n+1}, G)$.

Proof. For (a), observe that both vertices come from the same vertex in L^c_{n-1}, and so their distance in L^c_n is twice the distance to e^{n-1}_1 plus the length of the joining path, hence odd.

(b) follows from the fact that if $\mathcal{L}$ is large, it is not tiny. Hence, $\mathcal{L}(e^n_1)$ contains the endpoints of a minimal odd path of length $k \geq 2n + 1 > 2$, that is, there are $\varphi_0, \varphi_1 \in \mathcal{L}$ and a path in G $(\varphi_0(e^n_1) = v_0, \ldots, v_k = \varphi_1(e^n_1))$. Let $e_1, \ldots, e_k \in E$ be such that $\pi(e_j) = (v_{j-1}, v_j)$. Defining $d(n) = k - 2$ and $d(x) = c(x)$ for all $x \neq n$, notice that, by the remark preceding this lemma, $L^c_n = L^d_n$. I find $\varphi \in \mathrm{Hom}(L^d_{n+1}, G)$ be given by $\varphi^0 = \varphi_0$, $\varphi^1 = \varphi_1$ and $\varphi(p_i) = v_{i+1}$ for $0 \leq i \leq k - 2 = d(n)$; the edges are given by $\varphi(p_j, p_{j+1}) = e_j$ for $0 \leq j < k - 2$, $\varphi(e^n_1{}^\frown(0), p_0) = e_1$ and $\varphi(p_{k-2}, e^n_1{}^\frown(1)) = e_k$.

I prove (c) by contradiction. Recall that constructing L^c_{n+1} from L^c_n requires only knowing the value of $c(n)$. Suppose that for all d that are equal to c except in the n-th value, $\mathcal{L}^{+d} = \bigcup_m \mathcal{F}^d_m$ where each $\mathcal{F}^d_m$ is a tiny Borel subset of $\mathrm{Hom}(L^d_{n+1}, G)$. For each m and d, let $v^d_m \in V(L^d_n)$, $i^d_m < 2$ and $n_d(m)$ be the minimal natural such that $\Phi(\mathcal{F}^d_m(v^d_m{}^\frown(i^d_m)), m_d(n))$.

Now use the Reflection Lemma (reference **??** to be added) to find a Borel $B^d_m \supseteq \mathcal{F}^d_m(v_m^\frown(i_m))$ such that $\Phi(B^d_m, m_d(n))$. Consider $\mathcal{H}^d_m = \{\varphi \in \mathcal{L} : \varphi(v^d_m) \in B^d_m\}$, which is a Borel tiny subset of $\mathrm{Hom}(L^d_n, G) = \mathrm{Hom}(L^c_n, G)$. Indeed, notice that $\mathcal{H}^d_m(v^d_m) = \{\varphi(v^d_m) : \varphi \in \mathcal{H}^d_m\} = \{\varphi(v^d_m) : \varphi \in \mathcal{L} \wedge \varphi(v^d_m) \in B^d_m\} \subseteq B^d_m$.

Consider the large Borel set $\mathcal{L}_-$ obtained from removing from $\mathcal{L}$ the countable union of all tiny sets of the form $\mathcal{H}^d_m$. By part (b), there exist φ and d such that $\varphi \in \mathcal{L}^{+d}_- \subseteq \mathcal{L}^{+d} \subseteq \mathrm{Hom}(L^d_{n+1}, G)$. Suppose that $\varphi \in \mathcal{F}^d_m$, then $\varphi^{i^d_m}(v_m) = \varphi(v^d_m{}^\frown(i^d_m)) \in \mathcal{F}^d_m(v^d_m{}^\frown(i^d_m)) \subseteq B_m$, which means that $\varphi^{i^d_m} \in \mathcal{H}^d_m \cap \mathcal{L}_- = \emptyset$; a contradiction. □

4 Construction of a minimal graph of Borel chromatic number at least 3

The goal of this sections it to construct a family of graphs $\mathbb{L}_c$ indexed by reals $c \in \omega^\omega$.

Now, fix c and consider X_c as the set of all tuples $(n, k, x) \in \mathbb{N} \times \mathbb{N} \times 2^{\mathbb{N}}$ with $k \leq \max\{1, c(n)\}$ and notice that it is a closed subspace of the product space, and hence Polish. Define, for each (m, k, x) with $m \leq n$, $\pi_n(m, k, x)$ as the vertex of L^c_n determined by these parameters. Formally, $\pi_n(m, k, x) := (p_k)^\frown x \restriction (n - m)$. For example, $\pi_3(1, 2, 0110\cdots) = (p_2, 0)$ can be seen in Figure 1 labeled as $(2, 0)$.

Finally, $\mathbb{L}_c$ is the graph on X_c where two (n_i, k_i, x_i), for $i < 2$, are adjacent if and only if the $\pi_n(n_i, k_i, x_i)$ are adjacent in all L^c_n with $n \geq \max\{n_0, n_1\}$. Some basic properties of $\mathbb{L}_c$ follow.

Lemma 4.1.

(a) Two vertices $(n_i, k_i, x_i) \in X_c$, for $i < 2$, are in the same connected component of $\mathbb{L}_c$ if and only if there are $t_i \in 2^{<\omega}$ and $x \in 2^\omega$ such that $|t_0| - |t_1| = n_1 - n_0$ and $x_i = t_i^\frown x$.

(b) $\mathbb{L}_c$ has no cycles and all vertices of $\mathbb{L}_c$ have degree 2 except for the vertex $(0, 0, \{0\}^\omega)$, which has degree 1.

(c) $\chi_B(\mathbb{L}_c) = 3$.

Proof. (c) That $\chi_B(\mathbb{L}_c) \leq 3$ follows from Proposition 1.1 and the preceding fact. I argue the other inequality by contradiction. Suppose then that there is a partition of X into two independent sets. Since X is Polish, there is

a non-meager class B in the partition. B is comeager in some basic open set $[(n,k,t)]$. Skipping some technical details, this means that I can find a vertex $(n,k,t^\frown(\varepsilon)^\frown x) \in B$, and moreover not connected to any member of $[(n,k,t)] \setminus B$. I claim that $(n,k,t^\frown(1-\varepsilon)^\frown x) \in B$.

It then follows from Lemma 3.2(a) that these two vertices in B are an odd distance apart, which is impossible. □

The main result is then split into two parts.

Theorem 4.1. *For any analytic graph G, either $\chi_B(G) \leq 2$ or there is a c such that $\mathbb{L}_c \to_c G$.*

Moreover, c can be chosen to be unbounded and take only odd values.

Theorem 4.2. *Let $c, d \in \omega^\omega$ be unbounded and only taking odd values, then $\mathbb{L}_c$ and $\mathbb{L}_d$ are continuously homomorphically equivalent.*

5 Proof of Theorem 4.1

By Lemma 4.1(c) and Fact 1.1, it is clear that both conditions cannot hold simultaneously. Now suppose that G is an analytic graph of Borel chromatic number at least three. The theorem is proved by constructing a valid c and a continuous homomorphism witnessing $\mathbb{L}_c \to_c G$.

By Theorem 3.1, $V(G)$ is large. I recursively construct a function $c \in \omega^\omega$ and a sequence of large Borel sets $\mathcal{L}_n \subseteq \mathrm{Hom}(L_n^c, G)$ such that for all n,

1. $\mathcal{L}_{n+1} \subseteq \mathcal{L}_n^{+c}$;
2. for all $u \in V(L_n^c)$, the diameter of $\overline{\mathcal{L}_n(u)}$ is less than 2^{-n}; and
3. for all $(u,v) \in E(L_n^c)$, the diameter of $\overline{\mathcal{L}_n(u,v)}$ is less than 2^{-n}.

By (1), $\mathcal{L}_{n+1}(\pi_{n+1}(m+2,k,x)) \subseteq \mathcal{L}_n(\pi_n(m,k,x))$ for all $n > m$ and similarly for the edges: . For $(m,k,x) \in X_c$, let $f(m,k,x)$ be the unique point in

$$\bigcap_{n=m}^{\infty} \overline{\mathcal{L}_n(\pi_n(m,k,x))} = \bigcap_{n=0}^{\infty} \overline{\mathcal{L}_{n+m}((p_k)^\frown x \restriction n)}.$$

The map $f : X_c \to X$ is continuous.

It remains to check that it is a homomorphism from $\mathbb{L}_c$ to G. Suppose that $(m_i, k_i, x_i)_{i<2}$ are adjacent in $\mathbb{L}_0$, that is, for all $n \geq n_0 := \max\{m_0, m_1\}$, $\pi_n(m_i, k_i, x_i)_{i<2} \in L_n^c$. There is a unique edge $e \in E$ inside

$$\bigcap_{n \geq n_0} \overline{\mathcal{L}_n(\pi_n(m_0,k_0,x_0), \pi_n(m_1,k_1,x_1))}.$$

By continuity, $(f(m_0, k_0, x_0), f(m_1, k_1, x_1)) = \pi(e) \in G$. Thus the images form an edge of G.

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